

Nexivarin 10 mg Tablets — eCTD Submission
Module 3 — Quality

Submission Type	NDA — 505(b)(1)	Sequence Number	0000
Submission Date	2021-03-15	Section	3.2.S.2
Prepared by	PharmaCorp Inc.	Confidentiality	Proprietary and Confidential

Table of Contents — [3.2.S.2](#) Drug Substance — Manufacture

Section	Title	Description
3.2.S.2	Manufacture	Drug substance manufacturing overview
3.2.S.2.1	Manufacturer(s)	PharmaCorp API Division, Newark, NJ (DMF #32456); contract site Mumbai, India
3.2.S.2.2	Description of Manufacturing Process	5-step convergent synthesis route with process flow diagram
3.2.S.2.3	Control of Materials	Starting material specifications and supplier qualification
3.2.S.2.4	Controls of Critical Steps and Intermediates	Critical process parameters (CPPs) and in-process controls
3.2.S.2.5	Process Validation and/or Evaluation	Validation data for commercial-scale manufacturing

PharmaCorp Inc.
NDA 123456

Nexivarin 10 mg Tablets
Section 3.2.S.2
Drug Substance — Manufacture

Submission Type	NDA — 505(b)(1)
Sequence Number	0000
Submission Date	2021-03-15
Prepared by	PharmaCorp Inc.
Confidentiality	Proprietary and Confidential

3.2.S.2 Manufacture — Nexivarin Hydrochloride

3.2.S.2.1 Manufacturer(s)

Role	Facility	Address	DMF
API Manufacturer	PharmaCorp API Division	200 Production Blvd, Newark, NJ 07102	DMF #32456
Starting Material Supplier	Synthon Chemical Co.	45 Industrial Park, Mumbai, India 400703	Not applicable
Testing Laboratory	PharmaCorp QC Laboratory	200 Production Blvd, Newark, NJ 07102	—

3.2.S.2.2 Description of Manufacturing Process and Process Controls

Nexivarin hydrochloride is synthesized via a convergent 5-step route as summarized below. The process is performed under cGMP conditions per ICH Q7.

Step	Operation	Starting Material → Product	CPPs	CQAs
Step 1	Hantzsch Dihydropyridine Synthesis	2-Chlorobenzaldehyde + Ethyl acetoacetate + Aminoethoxymethyl acetoacetate → DHP Intermediate I	Temperature (60–70°C), Reaction time (8h), EtOH reflux	Yield (>75%), Impurity A <0.1%
Step 2	Selective Ester Hydrolysis	DHP Intermediate I → Half-acid Intermediate II	NaOH concentration, pH control (pH 10–11), Temperature (20–25°C)	Yield (>85%), residual NaOH <10 ppm
Step 3	Coupling & Cyclization	Intermediate II + ethanolamine derivative → Free base Intermediate III	Temperature, stoichiometry of coupling agent (EDC·HCl)	Impurity B <0.05%, residual EDC <50 ppm
Step 4	Crystallization (Free Base)	Intermediate III → Crystalline Free Base	Solvent ratio (EtOAc/Heptane 3:7), cooling rate (0.5°C/min)	Polymorphic form (Form I), particle size d90 <150 µm
Step 5	Salt Formation & Final Crystallization	Free Base + HCl/IPA → Nexivarin HCl (API)	HCl equivalents (1.02), temperature (0–5°C), seeding	Assay 98–102%, total impurities <0.3%, water <0.3%

3.2.S.2.3 Control of Materials

All starting materials, reagents, and solvents are controlled per approved specifications. Certificates of Analysis are obtained for each lot. Critical starting materials (CSMs) are 2-chlorobenzaldehyde (purity ≥99.0%) and the aminoethoxymethyl acetoacetate intermediate (purity ≥98.5%, sourced from approved suppliers). Elemental impurity controls are applied per ICH Q3D.

3.2.S.2.5 Process Validation and/or Evaluation

Three consecutive commercial-scale validation batches (800 kg each) were manufactured demonstrating consistent compliance with all API release specifications. CPP ranges were confirmed, and IPCs demonstrated process reproducibility. All validation batches were released for use in registration stability studies.